

Point Cloud Features-Based Random Forest for Road User Classification via Millimeter Wave Radar

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Abstract—Vehicle-mounted millimeter-wave(mmWave) radar sensors can meet the requirements of vehicles for the all-day and all-weather sensing of the surrounding environment. The mmWave radar can produce one to dozens of target points for ordinary road users, and such sparse point cloud data lead to difficulties in classification. When there are fewer or even only one point in the target-level point cloud, which usually occurs when the object is tiny or far from the radar, the classification accuracy of this type of object will become very poor. To improve the classification accuracy of radar-based road users, we propose a new feature extraction and hierarchical classification algorithm for mmWave radar point clouds. We extracted 6-dimensional and 125-dimensional features for single-point and multi-point objects based on the range, azimuth, velocity, and radar cross-section (RCS) of point clouds measured by the mmWave radar to characterize the motion, attitude, position, and material of the road user. To validate the effectiveness of the proposed method, we evaluate our method on the publicly available dataset. Compared with the traditional algorithm, the overall classification accuracy of the proposed method is higher. In addition, this divide-and-conquer strategy also improves the classification performance of single-point objects.

Keywords—feature extraction, radar point cloud, random forest, classification algorithms

I. INTRODUCTION

To effectively reduce traffic accidents and facilitate people's travel, the development of advanced driver assistance systems (ADAS) has received widespread attention worldwide. With the development of computer science, chip technology, artificial intelligence algorithms, various autonomous driving technologies have also made significant progress. Nowadays, ADAS systems integrate multiple sensors such as millimeter wave (mmWave) radar, LIDAR, and cameras. Compared with other commonly used sensors, mmWave radar has its unique advantages. On the one hand, mmWave radar is less affected by the environment. It can perform robust perception even in various harsh environments. In contrast, sooty or foggy environments can significantly affect lidar performance, and light and weather conditions can dramatically impact the imaging effect of the camera. On the other hand, mmWave radar is the only sensor that can obtain the precise radial velocity of all observed targets through a single measurement. In contrast, ordinary cameras and lidars cannot directly measure the highly accurate movement velocity of targets. The precise perception of velocity is necessary for many downstream tasks of autonomous driving.

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However, compared to other sensors, mainstream mmWave radar has the disadvantage of missing height information and lower angular resolution than other sensors. These drawbacks result in sparse data representations called radar point clouds for small objects at a distance, leading to difficulties in sensing based on mmWave radar point cloud data. For these reasons, the classification of road users based on mmWave radar data is emerging as both valuable and challenging research.

The mmWave radar can produce one to dozens of target points for ordinary road users from the Range-Doppler map or the Range-Azimuth-Doppler tensor because of its high range resolution and high velocity resolution. Compared with image-like radar data, the sparse point cloud data obtained by processing the image-like radar data occupies less computational and storage resources and is more suitable for autonomous driving scenarios with high real-time requirements and resource constraints. Based on the point-shaped echoes generated from Range-Doppler maps, Prophet R et al. manually extracted the number of multiple reflection points of the same object and its statistics of spread and variance in the range and velocity dimensions as features[1][2]. However, the above methods did not consider the information of the azimuth dimension of the object, and the performance was greatly affected by the object's location and orientation. Based on the point clouds from Range-Azimuth-Doppler tensors, Scheiner N et al. manually extracted 98-dimensional features and ensemble 21 Long Short-Term Memory (LSTM) classifiers to classify six classes of road users[3], and the method required data of duration 0.6s~0.8s to give the classification results. Zhao et al. extracted 11-dimensional features from the point clouds and used SVM classifiers to perform the binary classification for pedestrians and cars with good classification results in a simple experimental scenario[4]. Kentaro Toyoda et al. propose an on-road object identification considering time series of radar information and extract features from a series of time windows by calculating the mean and variance of object information[5].

Previous methods often suffer from some of the following shortcomings: 1) The methods were evaluated only in simple experimental scenarios; 2) The features extracted are so few that they do not adequately reflect the various characteristics of the object; 3) The algorithm consumes many computing resources and does not meet the real-time requirements of the autonomous driving system. In addition, we found in our experiments that single-point target classification is extremely poor, and this problem has not been focused on in previous research. Based on the above analysis, we propose a new

method for road user classification based on mmWave radar point clouds. The main contributions are as follows:

- 1) We propose a hierarchical classification framework to improve the classification accuracy of single-point objects by treating single-point objects and multi-point objects separately. To the best of our knowledge, there is no mention of mitigating the poor performance of single-point target classification in previous research.
- 2) We propose up to 125-dimensional feature vectors to characterize the shape, velocity, intensity, and position of the point cloud to describe the object entirely, and our method was evaluated in the publicly available realistic mmWave radar point cloud dataset.
- 3) We use probability vectors of multi-frame data for decision-level fusion to further improve classification performance without significant additional storage and computational resources and investigate the impact of decision fusion duration on classification performance.

Next, we will introduce the proposed method in Section II, give the experimental results and analysis in Section III, and conclude this paper in Section IV.

II. THE PROPOSED CLASSIFICATION METHOD

The radar point cloud can be obtained from the radar echo signal detected by the constant false alarm rate (CFAR) algorithm. Only the target reflection points above the adaptive amplitude threshold are retained. The radar point clouds are represented as a set of seven-dimensional points:

$$P = \{p_i = (x_i, y_i, \tilde{v}_i, \sigma_i, r_i, \theta_i, v_i) \mid i = 1, 2, \dots, n\} \quad (1)$$

where $n \in N$ denotes the number of radar detection points. Each point contains two cartesian coordinates (x, y) , two polar coordinates (r, θ) , ego motion compensated doppler velocity $\tilde{v}$, uncompensated doppler velocity v , and radar cross section values σ .

The conventional radar data classification process includes three steps as shown in Fig. 1: firstly, the object clusters are generated from the radar data by a clustering algorithm or the ground truth annotation, then the feature extraction is performed from the cluster, and finally classifying objects based on feature vectors which are extracted from the cluster. Extracting valuable features and choosing suitable classifiers are essential to produce good classification results. This paper focuses on improving classification accuracy by improving the second and third steps of the radar data classification process, and the cluster is grouped based on the ground truth provided in the dataset.



Fig. 1. Simplified processing chain for road user classification based on mmWave radar point cloud

A. Feature Extraction

For each cluster generated by the ground truth annotations, features based on $x, y, \tilde{v}, \sigma, r, \theta$ and v were created. For single-point clusters with one point and multi-point clusters with more than one point, we extracted 6-dimensional feature vectors and 125-dimensional feature vectors, respectively.

The six types of features we propose will be analyzed and elaborated on below.

1) The shape features of multi-point cluster

Based on the spatial coordinates in the radar measurement value, we extract the shape feature set about the cluster contour and size.

- The two eigenvalues and two squared eigenvalues of the covariance matrix of x and y , namely $xy\text{Lambda}1$, $xy\text{Lambda}2$, $xy\text{LambdaQuad}1$, and $xy\text{LambdaQuad}2$.
- The axis lengths of 95% confidence ellipses based on the covariance matrix of x and y , namely $majorLength$ and $minorLength$.
- The number of points, compactness, width of the cluster, namely $n\text{Detects}$, $Compactness$, and $clusterWidth$.
- The mean distance from each point of the cluster to the $clusterWidth$ vector, namely $maxDistDev$.
- The area, perimeter and density of the minimum bounding rectangle of the cluster, namely $areaRectHull$, $perimeterRectHull$, and $densityRectHull$.
- The area, perimeter, density and roundness of the convex hull of the cluster, namely $areaConvexHull$, $perimeterConvexHull$, $densityConvexHull$, and $Circularity$.
- The radius of the best fitting circle and the minimum bounding circle of the cluster, namely $radiusCircleFit$, and $radiusCircleMin$.
- The correlation coefficient of the x and y , namely $xyLinearity$.
- The average distance between the points of the cluster, namely $meanClusterWidth$.
- The spread, mean absolute deviation, variance, standard deviation, skewness, and kurtosis of x , namely $spreadX$, $madX$, $varX$, $stdX$, $skewX$, and $kurtX$.
- The spread, mean absolute deviation, variance, standard deviation, skewness, and kurtosis of y , namely $spreadY$, $madY$, $varY$, $stdY$, $skewY$, and $kurtY$.
- The area and density of the minimum bounding rectangle corresponding to the spread of x and y , namely $areaXY$ and $densityXY$.
- The spread of r and its square root value, logarithmic value, and squared value, namely $spreadRange$, $sqrRangeSpread$, $logRangeSpread$, and $quadRangeSpread$.
- The spread of θ and its square root value, logarithmic value, and squared value, namely $spreadAngle$, $sqrAngleSpread$, $logAngleSpread$, and $quadAngleSpread$.
- The spread, mean absolute deviation, variance, standard deviation, skewness, and kurtosis of r , namely $spreadRange$, $madRange$, $varRange$, $stdRange$, $skewRange$, and $kurtRange$.

- The spread, mean absolute deviation, variance, standard deviation, skewness, and kurtosis of θ , namely *spreadAngle*, *madAngle*, *varAngle*, *stdAngle*, *skewAngle*, and *kurtAngle*.

2) The velocity features of multi-point cluster

Based on the velocity $\tilde{v}$ and v , we extract the velocity feature set about the cluster velocity distribution. The following features were calculated:

- The minimum value, maximum value, spread, mean absolute deviation, variance, standard deviation, skewness, and kurtosis of $\tilde{v}$, namely *minVelocity*, *maxVelocity*, *spreadVelocity*, *madVelocity*, *varVelocity*, *stdVelocity*, *skewVelocity*, and *kurtVelocity*.
- The standard deviation of the uncompensated velocity v , namely *stdVelocityUncomp*.
- The mean value of $\tilde{v}$ and its square root value, logarithmic value, and squared value, namely *sqrMeanVelocity*, *logMeanVelocity*, and *quadMeanVelocity*.
- The histograms of $\tilde{v}$, namely *v1*, *v2*, *v3*, *v4*, and *v5*.

3) The intensity features of multi-point cluster

Based on the σ value, we extract the intensity feature set about the cluster intensity distribution.

- The minimum value, maximum value, spread, mean absolute deviation, variance, standard deviation, skewness, and kurtosis of σ , namely *minRCS*, *maxRCS*, *spreadRCS*, *madRCS*, *varRCS*, *stdRCS*, *skewRCS*, and *kurtRCS*.
- The mean value of σ and its square root, logarithmic value, and squared value, namely *sqrMeanRCS*, *logMeanRCS*, and *quadMeanRCS*.
- The histograms of σ , namely *rcs1*, *rcs2*, *rcs3*, *rcs4*, and *rcs5*.

4) The location features of multi-point cluster

Based on the spatial coordinates in the radar measurement value, we extract the location feature set about the position of the cluster relative to the radar.

- The minimum value, maximum value and mean value of r and θ , namely *minRange*, *maxRange*, *meanRange*, *minAngle*, *maxAngle*, and *meanAngle*.
- The distance and azimuth of the cluster relative to the radar, namely *meanDistance* and *meanOrientation*.
- The minimum value, maximum value and mean value of x and y , namely *minX*, *maxX*, *meanX*, *minY*, *maxY*, and *meanY*.

5) The combination of the basic units (x , y , $\tilde{v}$, σ , v , r and θ) of multi-point cluster

Some of the measured values of the radar alone cannot fully reflect the characteristics of the cluster, so we combined multiple basic units to extract the following features.

- The four eigenvalues and four squared eigenvalues of the covariance matrix of x , y , $\tilde{v}$ and σ , namely *xyvrLambda1*, *xyvrLambda2*, *xyvrLambda3*, *xyvrLambda4*, *xyvrLambdaQuad1*, *xyvrLambdaQuad2*, *xyvrLambdaQuad3*, and *xyvrLambdaQuad4*.

- The axis lengths of 95% confidence ellipses based on the covariance matrix of x , y , $\tilde{v}$ and σ , namely *axisLength1*, *axisLength2*, *axisLength3*, and *axisLength4*.

- The *spreadAngle* value and *nDetects* value weighted by *meanRange*, namely *spreadAngleComp* and *nDetectsComp*.

- The linear correlation between $\tilde{v}$ and r values, namely *rVrLinearity*.

- The linear correlation between $\tilde{v}$ and θ values, namely *angleVrLinearity*.

- The linear correlation between detection dilatation in major or minor direction of x/y confidence ellipses, namely *majorVrLinearity* and *minorVrLinearity*.

- The ratio of *spreadRange* and *spreadAngle* to *spreadVelocity*, namely *rVrSpread* and *angleVrSpread*.

- The ratio of spatial spread along major or minor axis of x/y confidence ellipse to *spreadVelocity*, namely *majorVrSpread* and *minorVrSpread*.

6) The characteristic of single-point cluster

When the cluster has only one point, we cannot extract its shape and statistical features, but we did not directly abandon the classification of these clusters, but only extracted the following features for subsequent classification tasks

- The basic units of radar measurements value x , y , $\tilde{v}$, σ .
- The combination of σ value and r value, namely σR .
- The combination of $\tilde{v}$ value and θ value, namely v_{back} .

B. Random Forest Classifier

Random forests alleviate the problem that single decision trees are prone to overfitting by introducing random attribute selection in the training process of decision trees. Compared with other commonly used classifiers, Random Forest has the following unique advantages: 1) Few hyperparameters, so there is almost no need to manually adjust; 2) Parallel operation, short training time; 3) Scale invariance to input features; 4) The importance of features can be output directly for feature selection; 5) Not sensitive to unbalanced data sets.

C. Training and Evaluation

Because our method extracts feature vectors of different dimensions for single-point clusters and multi-point clusters, it is necessary to train two different classifiers for clusters with different numbers of points. Before training, we first determined the hyperparameters of the random forest classifier by performing a grid search on the training set, including the number n_T of the trees, the depth d of the trees, and the number n_{split} of features used for splitting.

We will use the corresponding single-point or multi-point classifiers for classification for clusters with different numbers of points. Because the data imbalance is often faced when classifying road users, we use macroF1 as the evaluation metric of the classifier to avoid the classifier's preference for classes with a large number of samples.

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Introduction

RadarScenes[6] is a real-world radar point cloud data set for automotive applications. It consists of 5 large main categories, four radar sensors, and over 7000 road users, which are manually labeled on 100km of 158 different sequences. According to the official classification criteria of the dataset, 130 sequences were used as training data, and the other 28 sequences were used as test data. For feature extraction, all labeled cluster sequences are first sampled in time using a window of 180ms. The point clouds of the different time steps were transformed into the vehicle coordinate system at the time of the earliest measurement. The class membership of road users is distributed as depicted in Tab. I.

TABLE I SAMPLE DISTRIBUTION IN DATASET

Car	Large Vehicle	Two Wheel	Pedestrian	Pedestrian Group
131072	19313	64227	82268	24711

B. Comparison With Other Classification Method

To demonstrate the effectiveness of the proposed method, we compared it with a recent method[4] based on mmWave radar point clouds for road users classification.

TABLE II QUANTITATIVE RESULTS OF THE CLASSIFICATION METHOD BASED ON THE MILLIMETER WAVE RADAR POINT CLOUD

	Macro Precision	Macro Recall	Macro F1	Macro F1 (single-point clusters)
Proposed method	0.8539	0.8168	0.8316	0.4570
Feature set from Ref.[4]	0.7882	0.7481	0.7609	0.3842

We evaluate the overall classification performance in micro-precision, macro-recall, and macro-F1 in Table II for quantitative analysis. To verify the classification ability of the proposed method for single-point targets, we also evaluated macro-F1 under the single-point cluster test subset. We can conclude that our method has a better classification performance on the test sequence, proving that the proposed method significantly outperforms the comparison method.

C. Evaluation Of Feature Subsets

To better analyze the influence of various types of features on classification, we conducted classification experiments using different types of features separately.

TABLE III QUANTITATIVE RESULTS OF CLASSIFICATION METHODS BASED ON DIFFERENT FEATURE SUBSETS

Feature subsets	Shape features	Velocity features	Intensity features	Location features	Combination features
Macro F1	0.7450	0.5685	0.5275	0.6255	0.7515

It can be seen from Tab. III. that various types of features are separable. Among them, the shape features and the combination features are better for classification.

However, it is still obviously not as good as using all the features for classification, which proves the effectiveness of the various types of features extracted in this paper and the necessity of using them jointly for classification.

D. Evaluation Of decision-level fusion duration

To further improve the classification accuracy, we have performed decision-level fusion on the classification results of adjacent frames. The data frame whose time difference with the current frame is less than the fusion duration is the adjacent frame, and the mean vector of the predict vector of the adjacent frame is the current frame's final predict vector.

TABLE IV QUANTITATIVE RESULTS OF DECISION FUSION DURATION

Fusion duration(ms)	180	360	540	720
Macro F1	0.8316	0.8512	0.8579	0.8630

As the decision-level fusion duration increases, the classification performance improves, but the performance improvement margin decreases with time. We will classify each frame of data and store the prediction vector according to the fusion time. According to the system's real-time performance and classification accuracy requirements, we can flexibly choose the decision fusion time.

IV. CONCLUSION

In this paper, a new road user classification method based on point cloud features of mmWave radar is proposed. We have processed single-point clusters and multi-point clusters separately, and the extracted features and trained classifiers are more compatible with the clusters of the corresponding number of points. For various types of clusters, we have extracted up to 125 dimensions of different types of features. These features used in combination can describe the various characteristics of the cluster more comprehensively. The decision-level fusion of multi-frame data further improves the classification accuracy without significantly increasing the resource occupation. The experimental results on the public large-scale mmWave radar point cloud data set show the superiority of the proposed method.

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